

STAT 443: Time Series and Forecasting Final Report

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Section 1: Summary

Objective:

This study evaluates classical time-series models for forecasting Gold Futures closing prices. We tested whether other forecasting methods could outperform a simple Naïve baseline in a dynamic market. Performance was strictly measured using moving 1-step ahead forecasts on a holdout set to prevent data leakage.

We did a 70-30 split and compared four predictive frameworks:

1. Baselines: Historical Average and Persistence (Naïve) rules.
2. Exponential Smoothing (ETS)
3. ARIMA
4. ARIMAX: Incorporating lagged indicators (S&P 500, USD, 10-Year yields, and WTI Crude).

Results:

Model	Holdout RMSE
ARIMAX	48.822
Persistence (Naïve)	48.844
ARIMA	48.844
ETS	48.852

Conclusion:

The ARIMAX model marginally outperformed the Naïve model, though the performance gap was negligible (0.022). Because the holdout set surged into a massive monotone upward trend, the complex models almost completely lost their predictive edge, struggling to compete with the Naïve rule’s ability to instantly anchor to the most recent observation proved superior to complex models, which suffered by attempting to revert to historical means.

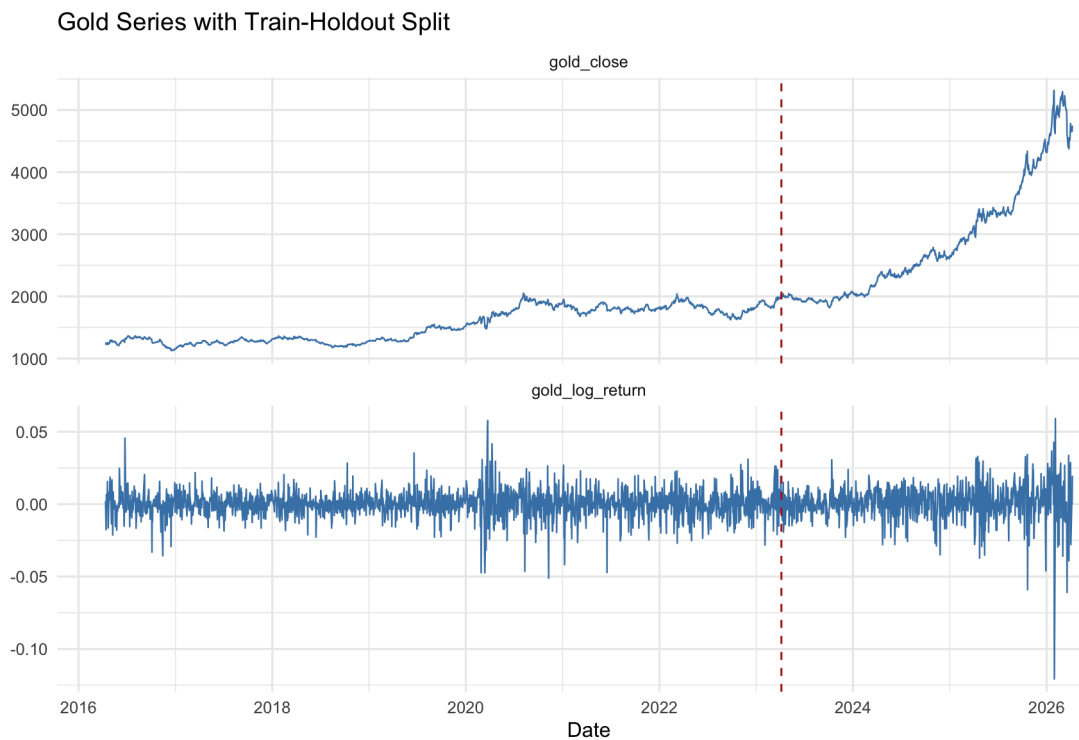
Section 2: Data Description

Data: Our analytical pipeline aggregates daily-frequency historical observations retrieved from Yahoo Finance and the Federal Reserve Economic Data (FRED).

To establish a continuous and mathematically uniform index, we utilized forward-fill imputation to reconcile holiday schedules and mismatched trading calendars.

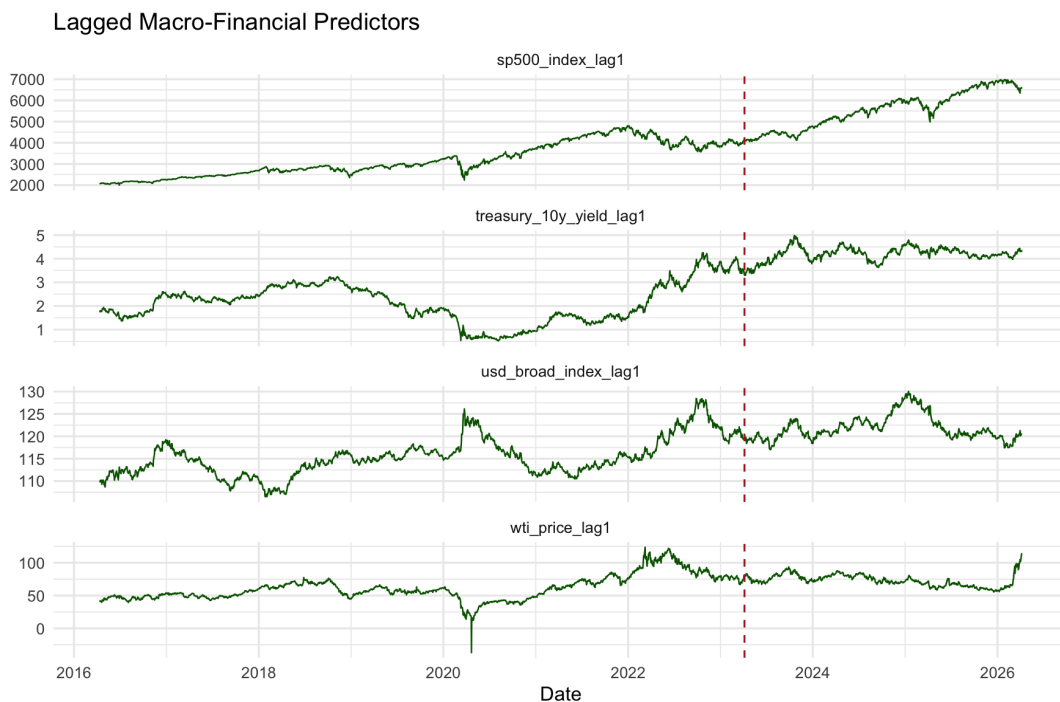
The Target Variable (Y_t): The primary dependent variable in this investigation is the closing price of [Gold Futures \(GC=F\)](#), denominated in USD per Troy Ounce. This asset serves as the definitive proxy for the daily market valuation and speculative consensus of gold.

- **Full Sample ($n = 2511$):** The aggregate historical series yields a mean closing price of 1929.665 with a standard deviation (SD) of 816.843.
- **Training Set ($n = 1757$):** During the training interval, gold prices demonstrated standard volatility patterns with a sample mean of 1533.160 (SD = 266.459).
- **Holdout Set ($n = 754$):** The out-of-sample block reveals a severe, monotone structural upward trend, with the sample mean shifting drastically to 2853.616 (SD = 914.811).



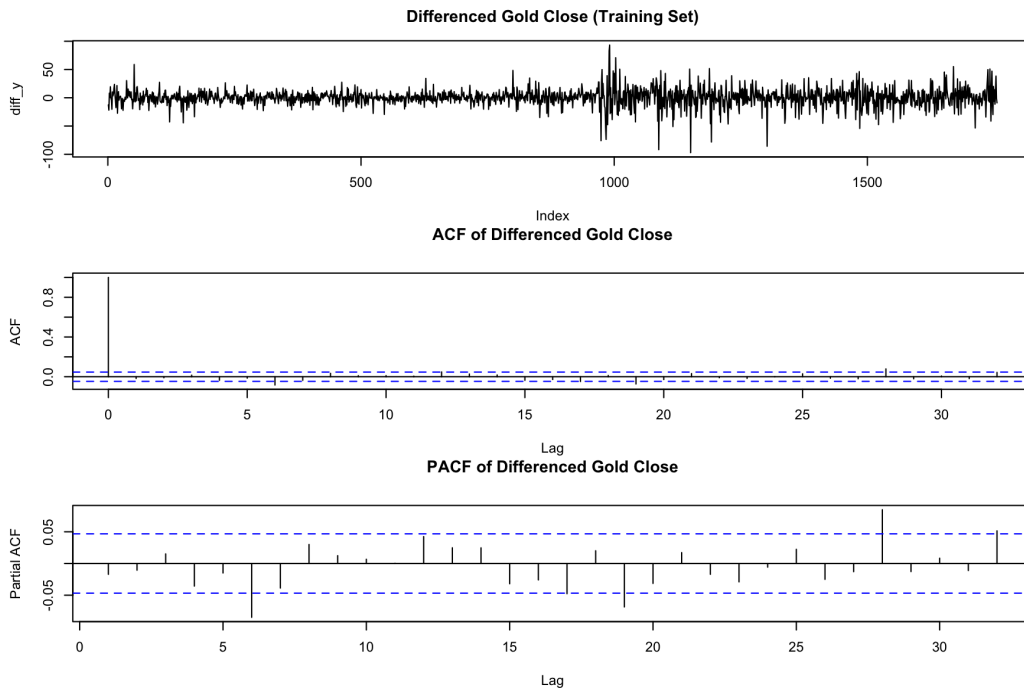
Macroeconomic Explanatory Variables (X_k): To parameterize our multivariate ARIMAX architecture, we selected a structurally relevant vector of four core macroeconomic indicators to capture systemic market pressures:

1. [U.S. Dollar Index \(DTWEXBGS\)](#) [X_1]: A broad index representing USD strength, which maintains a fundamentally robust inverse correlation with gold pricing. The full sample mean is 117.269 (SD = 4.863).
2. [S&P 500 Index \(SP500\)](#) [X_2]: Represents broad equity market performance and prevailing risk-on sentiment. The full sample mean is 3899.692 (SD = 1344.360).
3. [10-Year Treasury Yield \(DGS10\)](#) [X_3]: Denominated in percentages (%), this metric represents the risk-free rate and the direct opportunity cost of holding non-yielding assets. The full sample mean is 2.756 (SD = 1.192).
4. [WTI Crude Oil \(DCOILWTICO\)](#) [X_4]: Measured in USD per barrel, crude oil acts as a primary inflationary signal and a broad commodity-sector momentum indicator. The full sample mean is 64.861 (SD = 17.342).

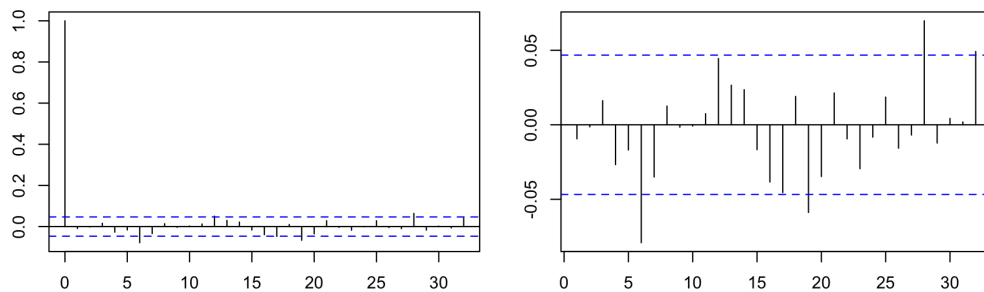


Mathematical Transformations: To satisfy the rigorous stationarity assumptions of classical time-series inference, we applied two strict mathematical transformations to the raw data prior to estimation:

1. **Logarithmic Returns:** Since non-stationary data invalidates ARIMA/ARIMAX assumptions, we used logarithmic returns ($r_t = \ln(y_t) - \ln(y_{t-1})$), yielding a full sample mean near zero at 0.001 (SD = 0.010).
2. **Strict 1-Day Exogenous Lags:** To prevent data leakage, all macroeconomic explanatory variables were strictly lagged by one period ($X_t^{lag} = X_{t-1}$). This structural design ensures that our **moving 1-step ahead forecasts** rely exclusively on information available prior to the observation day.



ACF and PACF respectively of Gold Log return (below)



Section 3: Analysis Outline

Establishing Theoretical Performance Floors: We established theoretical floors using two naive benchmarks: a historical Average rule and a Persistence (Naïve) rule. In dynamic financial markets that frequently exhibit random walk behaviors, the persistence rule (where $\hat{y}_{t+h|t} = y_t$) serves as a notoriously difficult baseline to defeat. As the holdout set transitions into a monotonically increasing trend from a training set of standard volatility, regression-based models often struggle against this rule, as naive anchoring adapts more rapidly to local mean shifts.

Exponential Smoothing (ETS): To capture the intrinsic smoothing parameters, we fitted parameterized exponential decay. Utilizing the `forecast` library, we recursively optimized against the training dataset to evaluate whether historical price action possessed exponentially decaying relevance.

ARIMA: Our approach to ARIMA modeling was rigorously diagnostic-driven, initiating the process with the `auto.arima()` algorithm to determine the optimal mathematical recommendations for auto-regressive (p), differencing (d), and moving average (q) orders.

Parameter selection and Validation: We manually adjusted the p and q lags to see if different configurations could produce better **1-step ahead** forecasts on our holdout data. To ensure the models were reliable, we checked the residuals (errors) using **ACF/PACF plots** and the **Ljung-Box test**. This confirmed the errors were **white noise**, proving the models captured all available patterns and left only random movement.

ARIMAX: To evaluate if global market constraints enhance predictive power, we expanded our univariate models into ARIMAX (ARIMA with Explanatory Variables). We strictly utilized a 1-day lagged vector of macroeconomic indicators, including Equities (S&P 500), U.S. Dollar strength, 10-Year Treasury Yields, and WTI Crude Oil.

Residual Modeling: This architecture was constructed by applying `auto.arima()` directly to the residual series to identify optimal structural errors. We then fitted multivariate regressions, integrating these structural errors with our lagged predictors and changing parameters to identify the configurations that generated the most precise moving 1-step ahead forecasts.

Strict Out-of-Sample Evaluation: The definitive selection of the topologically superior model relies exclusively on a strict evaluation framework, scoring competing methods across the hidden sequence using their generated Root Mean Square Error (RMSE) for the holdout set.

Computational Implementation of Forecasts: Since standard R libraries, such as `forecast::Arima()`, do not natively facilitate moving 1-step ahead holdout validation, we developed custom evaluation functions (`arma_fc()`, `arimax_fc()`). These functions utilize training estimates to sequentially update the data origin and calculate errors day-by-day.

Final Candidate Selection: Our final report filters out secondary iterations to compare exclusively the absolute best candidates from the ARIMA, ARIMAX, ETS, and Baseline classes, ranked strictly by their RMSE.

Section 4: Forecast Comparison

When comparing the moving 1-step ahead forecasts for the holdout set, the ARIMA class found its absolute best candidate in an ARIMA(2,1,2) with drift, which returned an RMSE of 48.844. By integrating our 1-day lagged macroeconomic variables, the multivariate ARIMAX class slightly improved upon this; the absolute best ARIMAX candidate (Regression with ARIMA(0,1,0) errors) achieved the lowest overall RMSE of 48.822. Our Exponential Smoothing model (ETS with multiplicative error, no trend, and no seasonality) achieved a very similar RMSE of 48.852.

Despite the mathematical complexity of these algorithms, when evaluating the moving 1-step ahead forecasts out-of-sample, the simple Persistence (Naïve) baseline fundamentally tied our best models, achieving a highly competitive holdout RMSE of 48.844. In drastic contrast, the historical Average baseline completely failed to compete, yielding a massive RMSE of over 1400.

Best Dynamic Regression Model: arimax_market_core_lag1
Regression with ARIMA(0,1,0) errors



Best ARIMA Model: auto_selected
ARIMA(0,1,0)



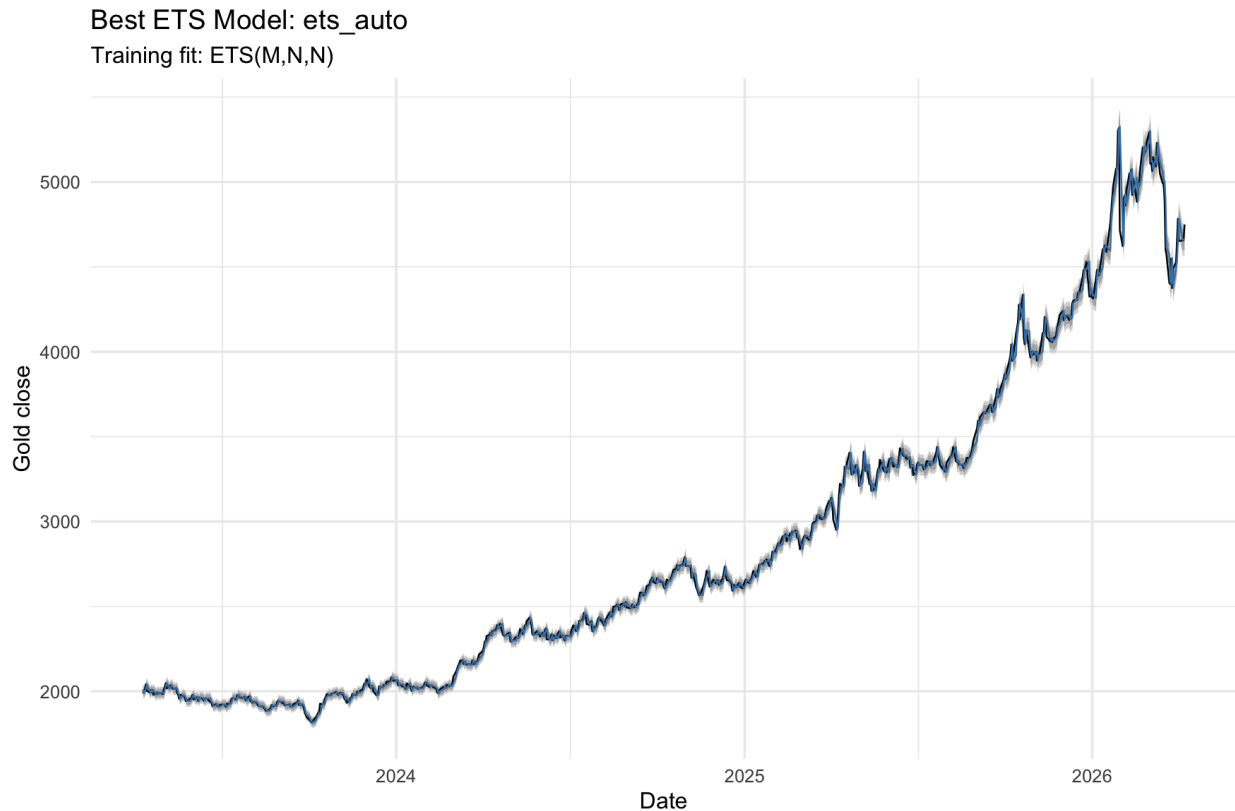
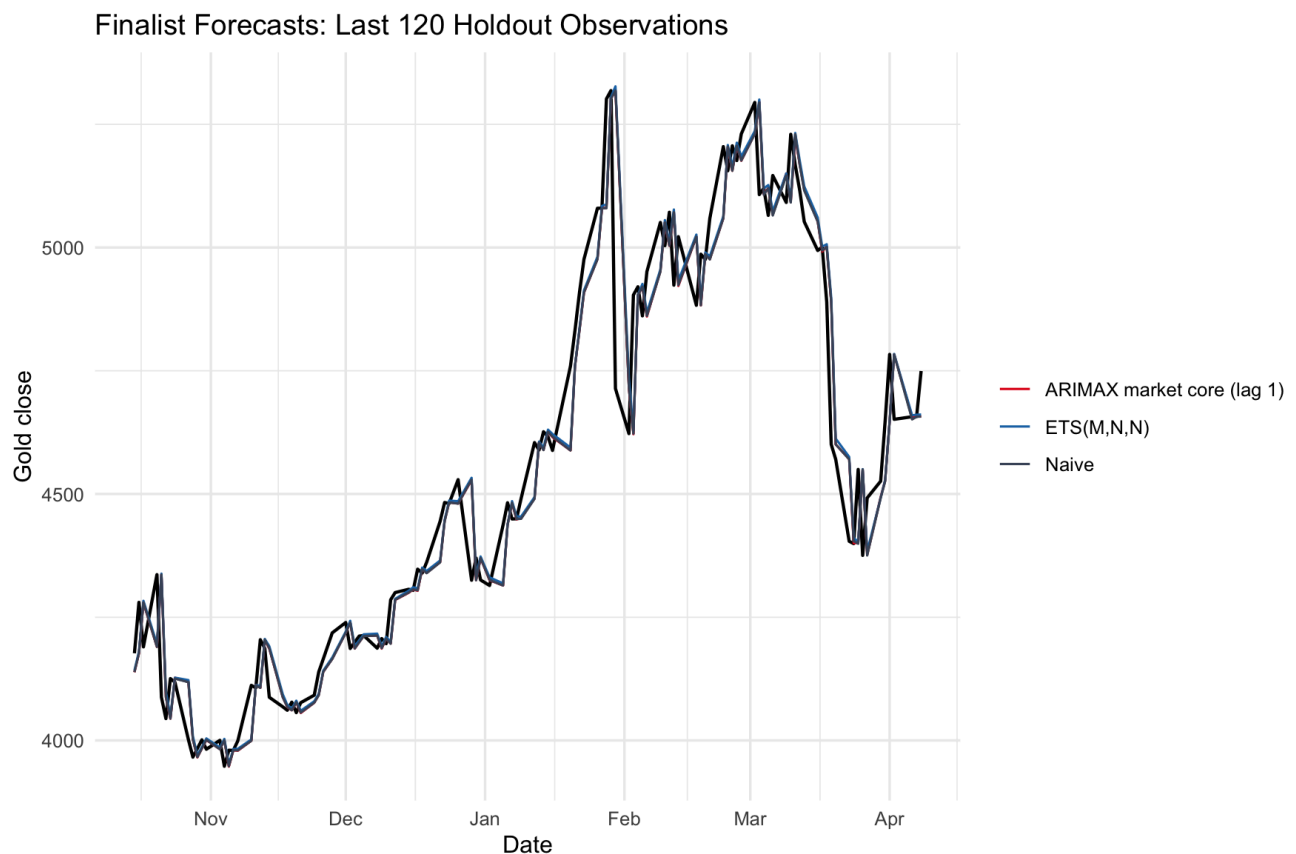


Table of Best Performing Models

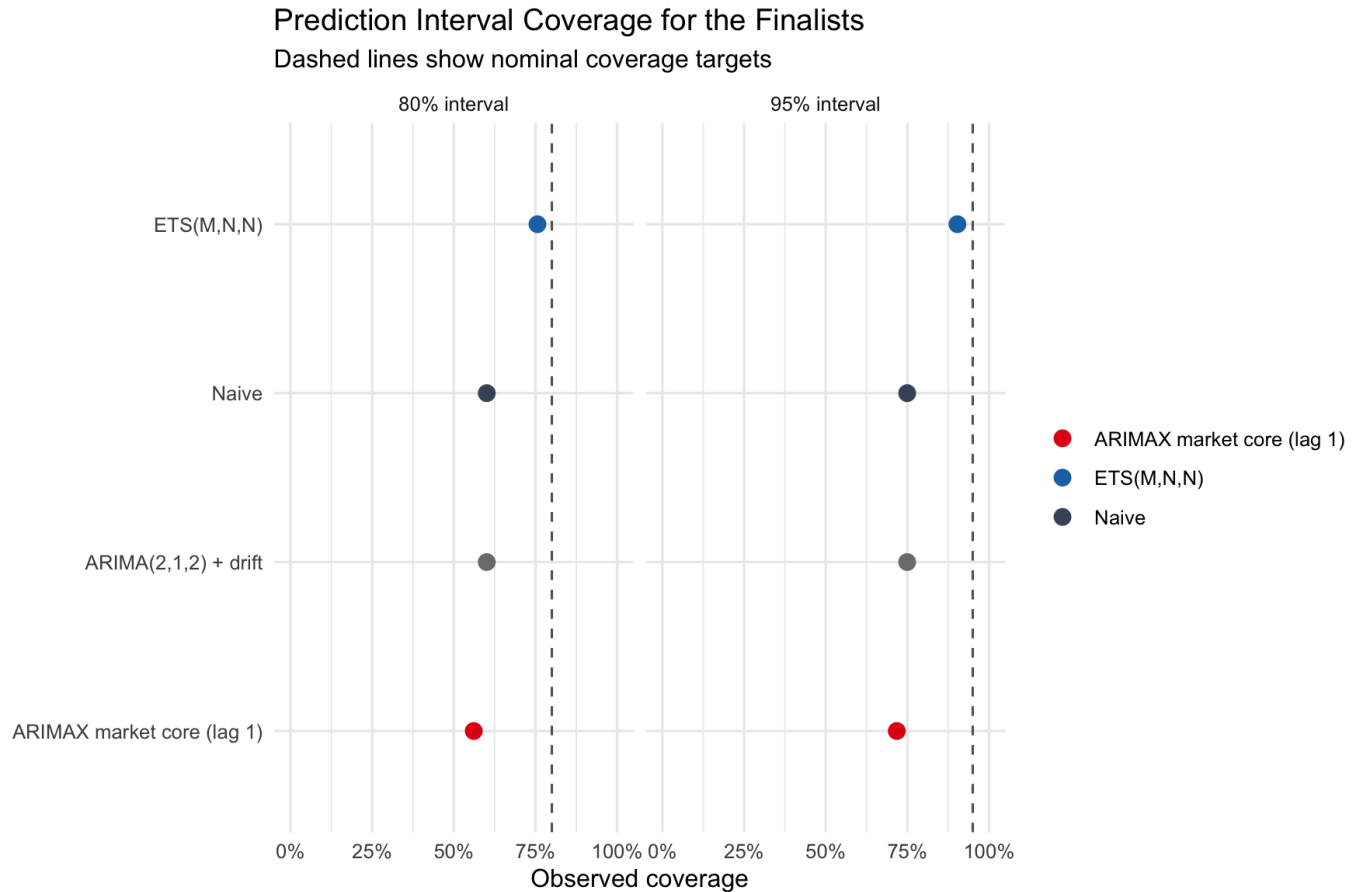
Model Class	Method	Holdout RMSE	80% P.I Coverage	95%P.I Coverage
ARIMAX	Regression with ARIMA(0,1,0) errors	48.822	0.561	0.718
Baseline	Naive benchmark	48.844	0.601	0.749
ARIMA	ARIMA(2,1,2) with drift	48.844	0.601	0.749
ETS	ETS(M,N,N)	48.852	0.756	0.903

While the multivariate ARIMAX model technically secured the lowest point-forecast error, the fact that the simple persistence rule effectively tied with the strongest mathematical models highlights a critical, naturally occurring statistical phenomenon in our dataset. The persistence rule behaves incredibly well when the training set exhibits standard periods of ups and downs, but the out-of-sample holdout set shifts into a severe, monotone trend.

Our summary statistics mathematically confirm this structural break. During the training sequence, the daily closing price of gold exhibited standard mean-reverting volatility with a sample mean of 1533.160. However, during the validation block, the asset transitioned into a massive, relentless monotone upward trend, pushing the holdout sample mean to 2853.616. In environments defined by a continuous monotone trend, the persistence rule establishes a notoriously difficult baseline to definitively beat. Because the naïve equation instantly anchors its prediction to the most recently observed value, it adapts to severe level shifts much faster than auto-regressive models, which often mathematically penalize themselves by attempting to pull predictions back toward historical norms.



Model Shortcomings and Suggested Improvements While the ARIMAX equation successfully won the global RMSE point-forecast competition, we must consider that our final forecast equation is not holistically good. Both ARIMAX and the naïve baseline struggled severely with interval forecasting. Our winning ARIMAX model achieved a 95% coverage rate of just 0.718, meaning actual prices fell completely outside of the model's 95% intervals nearly 28% of the time, proving the equation fails to appropriately model shifting variance. (For comparison, the purely endogenous ETS model managed a significantly more robust 95% coverage rate of 0.903).



To improve our multivariate ARIMAX equation and generate better explanatory power over out-of-sample variance, we must address the model's restrictive assumption of homoscedasticity. Standard ARIMA architectures assume that the variance of the error terms remains constant over time. However, our severe failure in prediction interval coverage mathematically proves that the gold holdout data exhibits heteroscedasticity i.e the volatility and variance dynamically shifted during the market run-up. To correct this, future iterations should transition to an ARIMAX-GARCH framework, which is specifically designed to model shifting variance and volatility clustering. In tandem with this architectural upgrade, we need exogenous variables uniquely tailored to market panic forecasting. Because gold is fundamentally a "safe-haven" asset, replacing nominal 10 year Treasury Yields with real interest rates (TIPS yields) would better measure the true opportunity cost of holding non-yielding bullion (gold or silver in bulk essentially). Finally, incorporating the Volatility Index (VIX) or a quantifiable Geopolitical Risk Index would serve as significantly better explanatory variables to predict the sudden, sharp demand spikes that caused the monotone trend during our holdout period.

Contributions

- 1) We previously mentored at a hackathon together.
- 2) Order: Alphabetically Surname

Team Member	Primary Responsibilities
Arav Dewan	Model fitting and Parameter estimation.
Saumya Jain (Leader)	Forecasting, Model fitting, and EDA plots.
Kashish Joshipura	Data cleaning, Validation, and Forecasting

Appendix

File name	Description
01_get_and_wrangle_data.R	Data preprocessing
02_split_and_EDA.R	Exploratory data analysis
03_baselines.R	Baseline estimates
04_ets.R	Exponential trend smoothing
05_arima.R	ARIMA
06_arimax.R	ARIMAX
07_compare_models.R	Model comparison

References:

[1]Cerqueira, V. (2025, January 19). *How to handle time series missing data*. Towards Data Science. <https://towardsdatascience.com/how-to-handle-time-series-missing-data-d45e9aaae72c/>

[2]Leveraging lagged exogenous variables for time-series forecasting – without time | by Tolga şakar | medium. (n.d.). <https://medium.com/@tlgskr/leveraging-lagged-exogenous-variables-for-time-series-forecasting-without-time-472f14acb488>

[3] Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice*. OTexts.

